

HDOC DAY – 9

QUESTION 1:

Write a menu-driven C program to input two numbers and swap them using a third variable & without using third variable. Also display the values before and after swapping.

Algorithm:

1. Start.
2. Input two integers a and b.
3. Display their values before swapping.
4. Display the swapping menu and read the choice.
5. For choice 1, use a third variable temp.
6. For choice 2, swap without a third variable.
7. Display values after swapping.
8. Stop.

Pseudocode:

READ a, b

PRINT a, b

DISPLAY menu

READ choice

IF choice = 1

temp = a

a = b

b = temp

ELSE IF choice = 2

a = a + b

b = a - b

$a = a - b$

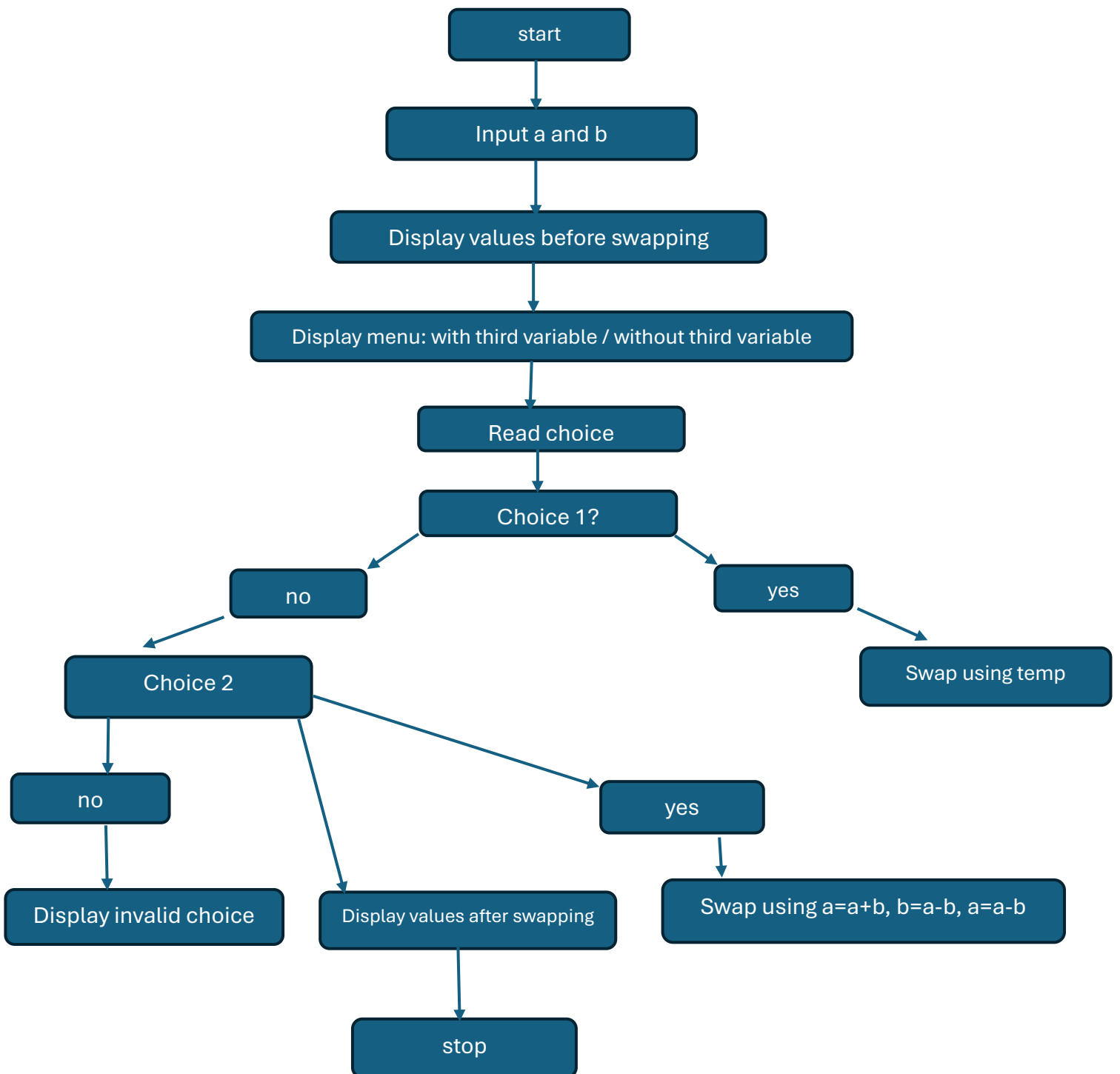
ELSE

PRINT "Invalid choice"

END IF

PRINT a, b

Flowchart:



C program:

```
#include <stdio.h>
```

```
int main() {
```

```
    int a, b, choice;
```

```
    printf("Enter two numbers: ");
```

```
    scanf("%d %d", &a, &b);
```

```
    printf("Before swapping: a = %d, b = %d\n", a, b);
```

```
    printf("\n1. Swap using third variable\n");
```

```
    printf("2. Swap without using third variable\n");
```

```
    printf("Enter your choice: ");
```

```
    scanf("%d", &choice);
```

```
    switch (choice) {
```

```
        case 1: {
```

```
            int temp = a;
```

```
            a = b;
```

```
            b = temp;
```

```
            break;
```

```
        }
```

```
        case 2:
```

```
            a = a + b;
```

```
b = a - b;
```

```
a = a - b;
```

```
break;
```

```
default:
```

```
printf("Invalid choice!\n");
```

```
return 0;
```

```
}
```

```
printf("After swapping: a = %d, b = %d\n", a, b);
```

```
return 0;
```

```
}
```

```
#include <stdio.h>

int main() {
    int a, b, choice;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("Before swapping: a = %d, b = %d\n", a, b);

    printf("\n1. Swap using third variable\n");
    printf("2. Swap without using third variable\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1: {
            int temp = a;
            a = b;
            b = temp;
            break;
        }

        case 2:
            a = a + b;
            b = a - b;
            a = a - b;
            break;

        default:
            printf("Invalid choice!\n");
            return 0;
    }

    printf("After swapping: a = %d, b = %d\n", a, b);

    return 0;
}
```

Test case table:

Test case	a.b	choice	Expected result
1	10, 20	1	After swap: a=20, b=10
2	-5, 8	1	After swap: a=8, b=-5
3	0,99	2	After swap: a=99, b=0
4	15,15	2	After swap: a=15, b=15
5	-12, -7	2	After swap: a=-7, b=-12

```

kali@kali:~$ ./h9
Enter two numbers: 10 20
Before swapping: a = 10, b = 20

1. Swap using third variable
2. Swap without using third variable
Enter your choice: 1
After swapping: a = 20, b = 10

kali@kali:~$ ./h9
Enter two numbers: -5 8
Before swapping: a = -5, b = 8

1. Swap using third variable
2. Swap without using third variable
Enter your choice: 1
After swapping: a = 8, b = -5

kali@kali:~$ ./h9
Enter two numbers: 0 99
Before swapping: a = 0, b = 99

1. Swap using third variable
2. Swap without using third variable
Enter your choice: 2
After swapping: a = 99, b = 0

kali@kali:~$ ./h9
Enter two numbers: 15 15
Before swapping: a = 15, b = 15

1. Swap using third variable
2. Swap without using third variable
Enter your choice: 2
After swapping: a = 15, b = 15

kali@kali:~$ ./h9
Enter two numbers: -12 -7
Before swapping: a = -12, b = -7

1. Swap using third variable
2. Swap without using third variable
Enter your choice: 2
After swapping: a = -7, b = -12

```

QUESTION 2:

Write a menu-driven C program to find the sum of the first n natural numbers and sum of squares of the first n natural numbers.

Algorithm

1. Start.
2. Input positive integer n.
3. Display menu and read choice.
4. If choice is 1, add numbers from 1 to n.
5. If choice is 2, add squares from 1^2 to n^2 .

6. Display the result.

7. Stop.

Pseudocode:

READ n

IF $n \leq 0$

 PRINT "Invalid input"

 STOP

END IF

DISPLAY menu

READ choice

sum = 0

IF choice = 1

 FOR i = 1 TO n

 sum = sum + i

 END FOR

 PRINT sum

ELSE IF choice = 2

 FOR i = 1 TO n

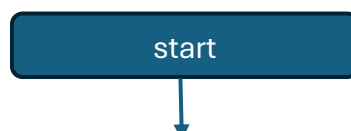
 sum = sum + $i * i$

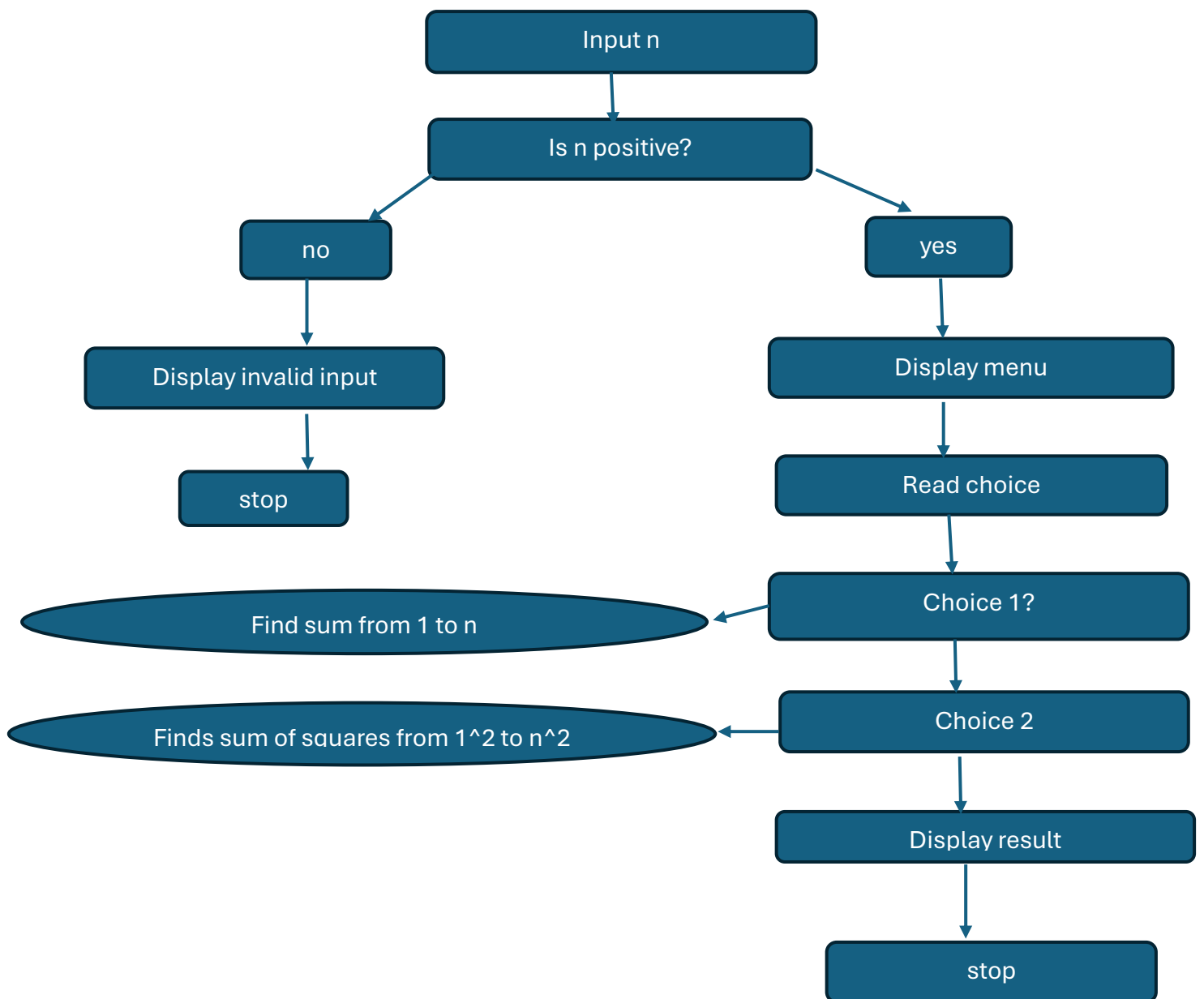
 END FOR

 PRINT sum

END IF

Flowchart:





C program:

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, choice, i;
```

```
    long long sum = 0;
```

```
    printf("Enter a positive integer n: ");
```

```
scanf("%d", &n);
```

```
if (n <= 0) {  
    printf("Invalid input! Enter a positive integer.\n");  
    return 0;  
}
```

```
printf("\n1. Sum of first n natural numbers\n");  
printf("2. Sum of squares of first n natural numbers\n");  
printf("Enter your choice: ");  
scanf("%d", &choice);
```

```
switch (choice) {  
    case 1:  
        for (i = 1; i <= n; i++) {  
            sum = sum + i;  
        }  
        printf("Sum of first %d natural numbers = %lld\n", n, sum);  
        break;
```

```
    case 2:  
        for (i = 1; i <= n; i++) {  
            sum = sum + (long long)i * i;  
        }  
        printf("Sum of squares of first %d natural numbers = %lld\n", n, sum);  
        break;
```

```
    default:
```

```

        printf("Invalid choice!\n");
    }

    return 0;
}

```

```

#include <stdio.h>

int main() {
    int n, choice, i;
    long long sum = 0;

    printf("Enter a positive integer n: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Invalid input! Enter a positive integer.\n");
        return 0;
    }

    printf("\n1. Sum of first n natural numbers\n");
    printf("2. Sum of squares of first n natural numbers\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            for (i = 1; i <= n; i++) {
                sum = sum + i;
            }
            printf("Sum of first %d natural numbers = %lld\n", n, sum);
            break;

        case 2:
            for (i = 1; i <= n; i++) {
                sum = sum + (long long)i * i;
            }
            printf("Sum of squares of first %d natural numbers = %lld\n", n, sum);
            break;

        default:
            printf("Invalid choice!\n");
    }

    return 0;
}

```

Test case table:

Test case	n	choice	Expected result
1	5	1	Sum = 15
2	10	1	Sum = 55
3	5	2	Sum of squares = 55
4	10	2	Sum of squares = 385
5	1	2	Sum of squares = 1

```

$ ./h92
Enter a positive integer n: 5

1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
Enter your choice: 1
Sum of first 5 natural numbers = 15

(kali@kali)-[~]
$ ./h92
Enter a positive integer n: 10

1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
Enter your choice: 1
Sum of first 10 natural numbers = 55

(kali@kali)-[~]
$ ./h92
Enter a positive integer n: 5

1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
Enter your choice: 2
Sum of squares of first 5 natural numbers = 55

(kali@kali)-[~]
$ ./h92
Enter a positive integer n: 10

1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
Enter your choice: 2
Sum of squares of first 10 natural numbers = 385

(kali@kali)-[~]
$ ./h92
Enter a positive integer n: 1

1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
Enter your choice: 2
Sum of squares of first 1 natural numbers = 1

```

QUESTION 3:

Write a menu-driven C program to calculate simple interest, compound amount, compound interest, and total payable amount for given principal, rate, and time.

Algorithm

1. Start.
2. Input principal P, rate R, and time T.
3. Calculate:
 - $SI = (P \times R \times T) / 100$
 - $A = P \times (1 + R/100)^T$

- $CI = A - P$

4. Display menu and read choice.

5. Display the selected result.

6. Stop.

Pseudocode:

READ P, R, T

$$SI = (P * R * T) / 100$$

$$A = P * (1 + R / 100)^T$$

$$CI = A - P$$

DISPLAY menu

READ choice

IF choice = 1

 PRINT SI

ELSE IF choice = 2

 PRINT A

ELSE IF choice = 3

 PRINT CI

ELSE IF choice = 4

 PRINT P + SI

 PRINT A

ELSE

 PRINT "Invalid choice"

END IF

C program:

```
#include <stdio.h>
```

```
#include <math.h>
```

```
int main() {
```

```
    double principal, rate, time;
```

```
    double simpleInterest, compoundAmount, compoundInterest;
```

```
    int choice;
```

```
    printf("Enter principal amount: ");
```

```
    scanf("%lf", &principal);
```

```
    printf("Enter rate of interest: ");
```

```
    scanf("%lf", &rate);
```

```
    printf("Enter time in years: ");
```

```
    scanf("%lf", &time);
```

```
    if (principal <= 0 || rate < 0 || time <= 0) {
```

```
        printf("Invalid input!\n");
```

```
        return 0;
```

```
    }
```

```
    simpleInterest = (principal * rate * time) / 100;
```

```
    compoundAmount = principal * pow((1 + rate / 100), time);
```

```
    compoundInterest = compoundAmount - principal;
```

```
    printf("\n1. Simple Interest\n");
```

```
    printf("2. Compound Amount\n");
```

```
    printf("3. Compound Interest\n");
```

```
printf("4. Total Payable Amount\n");  
printf("Enter your choice: ");  
scanf("%d", &choice);  
  
switch (choice) {  
    case 1:  
        printf("Simple Interest = %.2lf\n", simpleInterest);  
        break;  
  
    case 2:  
        printf("Compound Amount = %.2lf\n", compoundAmount);  
        break;  
  
    case 3:  
        printf("Compound Interest = %.2lf\n", compoundInterest);  
        break;  
  
    case 4:  
        printf("Total payable with Simple Interest = %.2lf\n",  
            principal + simpleInterest);  
        printf("Total payable with Compound Interest = %.2lf\n",  
            compoundAmount);  
        break;  
  
    default:  
        printf("Invalid choice!\n");  
}
```

return 0;

}

```
#include <stdio.h>
#include <math.h>

int main() {
    double principal, rate, time;
    double simpleInterest, compoundAmount, compoundInterest;
    int choice;

    printf("Enter principal Amount: ");
    scanf("%lf", &principal);

    printf("Enter rate of interest: ");
    scanf("%lf", &rate);

    printf("Enter time in years: ");
    scanf("%lf", &time);

    if (principal <= 0 || rate < 0 || time <= 0) {
        printf("Invalid input!\n");
        return 0;
    }

    simpleInterest = (principal * rate * time) / 100;
    compoundAmount = principal * pow((1 + rate / 100), time);
    compoundInterest = compoundAmount - principal;

    printf("\n1. Simple Interest\n");
    printf("2. Compound Amount\n");
    printf("3. Compound Interest\n");
    printf("4. Total Payable Amount\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            printf("Simple Interest = %.2lf\n", simpleInterest);
            break;

        case 2:
            printf("Compound Amount = %.2lf\n", compoundAmount);
            break;

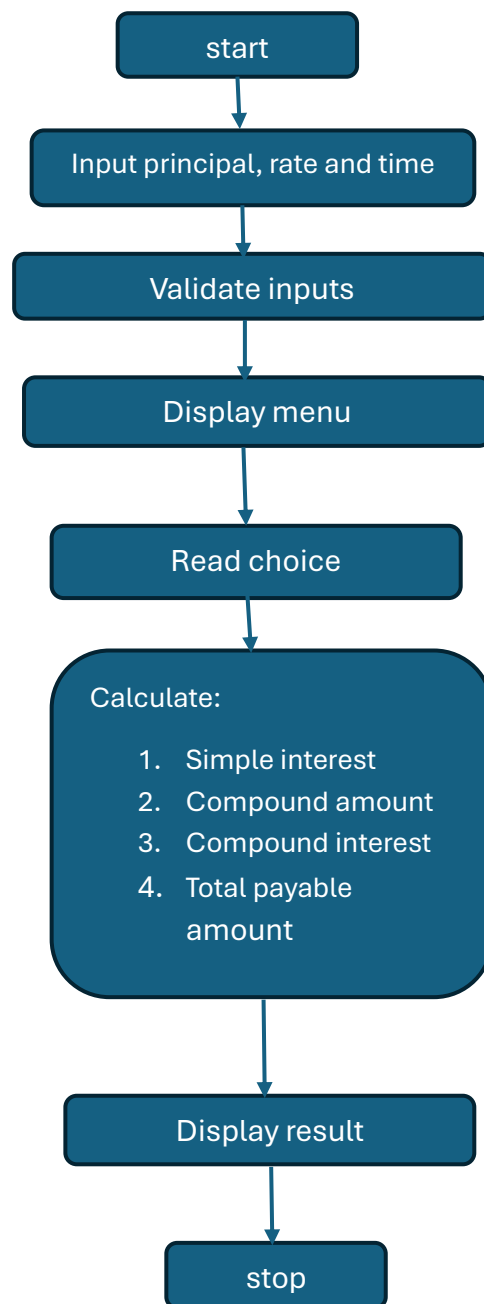
        case 3:
            printf("Compound Interest = %.2lf\n", compoundInterest);
            break;

        case 4:
            printf("Total payable with Simple Interest = %.2lf\n",
                principal + simpleInterest);
            printf("Total payable with Compound Interest = %.2lf\n",
                compoundAmount);
            break;

        default:
            printf("Invalid choice!\n");
    }

    return 0;
}
```

Flowchart:



Test case table:

Test case	principal	rate	time	choice	Expected result
1	10000	5%	2 years	1	Simple interest = 1000
2	10000	5%	2 years	2	Compound amount=11025

3	10000	5%	2 years	3	Compound interest=1025
4	10000	5%	2 years	4	Simple total = 11000 Compound total = 11025

```

(kali㉿kali)-[~]
└─$ ./h93
Enter principal amount: 10000
Enter rate of interest: 5
Enter time in years: 2

1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 1
Simple Interest = 1000.00

(kali㉿kali)-[~]
└─$ ./h93
Enter principal amount: 10000
Enter rate of interest: 5
Enter time in years: 2

1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 2
Compound Amount = 11025.00

(kali㉿kali)-[~]
└─$ ./h93
Enter principal amount: 10000
Enter rate of interest: 5
Enter time in years: 2

1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 3
Compound Interest = 1025.00

(kali㉿kali)-[~]
└─$ ./h93
Enter principal amount: 10000
Enter rate of interest: 5
Enter time in years: 2

1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
Enter your choice: 4
Total payable with Simple Interest = 11000.00
Total payable with Compound Interest = 11025.00

```